
What Does Discussion Teach? Behavioral and Mechanistic Effects of Multi-Agent RLVR at Matched Compute

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Abstract

Multi-agent discussion is a popular addition to RLVR training pipelines, but it’s rarely separated from the extra compute and reward-formula effects it introduces. We train five Qwen3-4B-Instruct models from the same base at identical compute: S (solo), D (CRISP-style discussion), C (a self-draft compute-matched control), DPP (strict-gate discussion), and CAP v2 (a single-agent control that delivers the persuader bonus’s gradient signal without any peer interaction). C beats D on accuracy on the hardest out-of-distribution evaluations, and D’s adversarial flip rate under wrong counter-arguments is the single behavioral fingerprint no other condition fully reproduces (roughly $6\times$ lower than S’s, $4\times$ lower than C’s). CAP v2 reaches an adversarial flip rate of 0.024 compared to D’s 0.012 with no peer mechanism, reproducing most of the fingerprint through solo training alone. A mechanism analysis using contrastive activation directions, with causal validation via steering on D, identifies a volume-versus-purity tradeoff in reward design: the reflection axis (causally responsible for D’s adversarial robustness) requires gradient volume on confident-correct rollouts, the commitment axis benefits from per-firing purity, and the standard persuader bonus gates both with the same trigger. D, DPP, and CAP v2 train different combinations of the two axes. The multi-agent setup’s contribution to adversarial robustness is mostly reproducible through single-agent gradient amplification; which residual mechanism the peer interaction adds is left open.

1 Introduction

Multi-agent discussion has become a popular addition to RLVR training. Systems like CRISP [Baumgartner and Agrawal, 2026], MAPoRL [Park et al., 2025], Dr. MAS [Feng et al., 2026], and Stronger-MAS [Zhao et al., 2026] report accuracy gains from peer interaction, evaluated against a solo-RLVR baseline trained on the same problems under the same algorithm. That comparison mixes peer interaction with the extra compute the discussion phase consumes and the gradient-shaping signal the multi-agent reward formula introduces. A multi-agent win against this baseline is consistent with any of the three driving the gain. The inference-time literature has already worked through the analogous argument: Wang et al. [2024], Becker et al. [2026], and Wynn et al. [2025] find single-agent controls at matched compute match or beat multi-agent protocols. This paper asks the analogous question during training, on the CRISP family.

The investigation trains five Qwen3-4B-Instruct models from the same base via GRPO on DAPO-Math-17K at identical compute. S is standard solo training, D is CRISP-style discussion, C is a self-draft compute-matched control that spends D’s extra compute on additional solo rollouts. Two follow-ups extend the analysis: DPP, a strict-gate discussion variant motivated by D’s mutual-swap

limitation, and CAP v2, a single-agent control that delivers the persuader bonus’s gradient signal via an advantage multiplier on confident-correct rollouts. The mechanism analysis identifies a volume-versus-purity tradeoff in reward design: the reflection axis (causally responsible for D’s adversarial robustness) requires gradient volume on confident-correct rollouts; the commitment axis benefits from clean per-firing signal; the standard persuader-bonus gates both with the same trigger.

Contributions.

- A matched-compute investigation of CRISP-family multi-agent RLVR with five trained conditions, and two analytical controls (self-draft compute-matched and confidence-amplified single-agent) developed during the investigation.
- A multi-axis mechanistic characterization (commitment, reflection, peer-framing) across all conditions, with causal validation of the reflection axis on D via activation steering.
- A volume-versus-purity tradeoff in reward design: the reflection axis requires gradient volume while the commitment axis benefits from purity, and the standard persuader-bonus gates both with the same trigger. CAP v2 partially confirms the volume hypothesis without any peer mechanism.

2 Setup

2.1 Training conditions

The investigation produces six trained conditions plus a base reference. The three primary analytical conditions are S, D, and C; DPP and CAP v2 extend the analysis; D-prime is an intermediate stepping stone appearing only in the mutual-swap progression in §3. All trained conditions share base weights (Qwen3-4B-Instruct-2507), data (DAPO-Math-17K filtered to 2,023 problems), GRPO setup, optimizer, learning-rate schedule, and total compute (200 iterations on $2 \times H200$). The variable that differs is what happens during the rollout phase. Hyperparameters are in Appendix A; iter 100 is analyzed throughout (iter 75 for D-prime).

base is untrained Qwen3-4B-Instruct-2507, used as a frozen reference for direction extraction and the KL anchor. **S (Solo)** is standard single-agent GRPO [Guo et al., 2025]: 8 rollouts per problem, no persuader bonus, no discussion stage, the baseline most multi-agent RLVR systems compare against. **D (Discussion)** uses two players, Alice at $T=0.6$ and Bob at $T=1.0$, following the CRISP discussion-only design [Baumgartner and Agrawal, 2026]: when their majority votes disagree, they exchange representative rollouts under a discussion prompt and generate revised solutions; GRPO updates Alice’s weights from the post-discussion rollouts, and a persuader bonus of $\gamma = 0.3$ is added to the correct pre-discussion rollouts of a player whose peer adopted their answer. **C (Compute Control)** keeps D’s two-player setup and disagreement detection but on disagreement each player generates one additional rollout from a self-draft prompt structurally identical to D’s, except both solutions shown are from the same player. D and C spend the same compute per disagreement event ($\sim 6k$ -token prefill, 2,048-token decode cap); the only difference is peer interaction vs additional solo sampling.

D-prime was a first attempt at addressing D’s mutual-swap limitation, adding four interventions: heterogeneous Bob (Qwen3-4B-Geo, an SFT of Instruct on AoPS-AMC), per-player KL anchors, a stability bonus of $\beta = 0.3$, and an XOR-disagreement curriculum filter. It reduced mutual swap but the persuader bonus still fired on mutual-swap events, motivating DPP. **DPP** keeps D-prime’s interventions and tightens the persuader gate: the bonus fires only when the persuader holds position post-discussion. **CAP v2** is solo training with a per-rollout advantage multiplier of $(1 + \gamma)=1.3$ applied at the loss layer to correct rollouts on problems the model already solves confidently ($\geq 5/8$ of 8 rollouts correct). The multiplier acts after GRPO’s per-group z-scoring; rewards are untouched, the running baseline does not drift. The CAP v1 reward-additive precursor collapsed at iter 80 due to baseline drift, motivating the v2 redesign (§4.2).

2.2 Evaluation

Pass@1 is measured at $T=0.6$ with 8 rollouts per problem on MATH-500 (L1–L5) [Hendrycks et al., 2021], GPQA Diamond [Rein et al., 2024], and AIME 2024/25; L5 is additionally evaluated at

Table 1: Primary analytical conditions. Each row names a trained condition and the components beyond standard single-agent GRPO that distinguish it. D-prime appears only in the mutual-swap progression in §3 and is omitted.

Condition	What it adds
S	Standard single-agent GRPO. No additions.
D	Two players (Alice $T=0.6$, Bob $T=1.0$). Discussion on disagreement. Loose persuader bonus ($\gamma=0.3$).
C	Two players. Self-draft extra rollout on disagreement instead of discussion. No persuader bonus.
DPP	Two players, heterogeneous Bob (Qwen3-4B-Geo via SFT), per-player KL anchors, stability bonus ($\beta=0.3$), XOR curriculum filter, strict persuader gate.
CAP v2	Single-agent. Advantage multiplier $1.3\times$ on correct rollouts of confident-correct problems ($\geq 5/8$). No peer, no discussion.

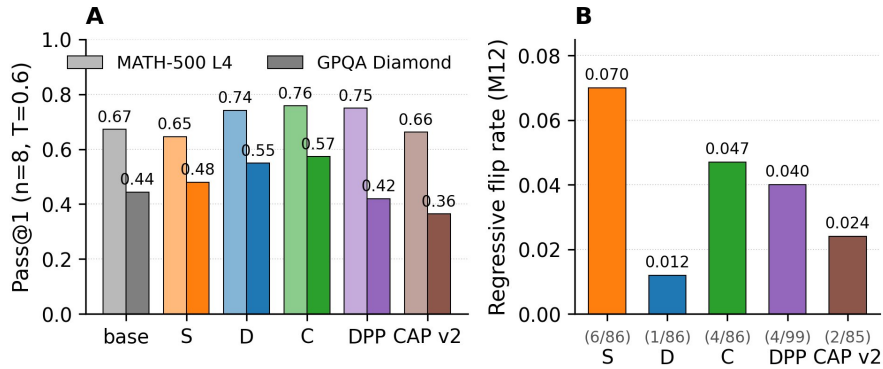


Figure 1: Behavioral fingerprint across five trained conditions plus base. **A:** Pass@1 on MATH-500 L4 ($n=8$, $T=0.6$) and GPQA Diamond ($n=8$, $T=0.6$). C wins both benchmarks; D and DPP track close behind. **B:** Regressive flip rate (M12) on the include-yourself intersection. D is roughly $6\times$ lower than S and $4\times$ lower than C. CAP v2 (no peer mechanism) reaches 0.024, reproducing most of D’s adversarial-robustness fingerprint through solo training.

$n=256$ for pass@k. Headline capability claims rest on MATH-500 L4 ($n=128$) and GPQA Diamond ($n=198$); L1–L3 are tied within 4–5pp of base across conditions, AIME 24/25 are too small ($n=30$ each) for strong claims.

Adversarial robustness is the regressive flip rate: on problems the model solves confidently unsteered, we append a plausible wrong counter-argument to its correct response and record the fraction of trials on which the model abandons its initial answer. The intersection set follows an include-yourself convention: each condition’s flip rate is computed where every condition in the comparison group, including itself, solves at least $5/8$ unsteered rollouts. This gives $n=86$ for S, D, C; $n=99$ for DPP; $n=85$ for CAP v2. Cross-condition comparisons disclose the denominator inline. Probing uses residual-stream activations at layers $\{0, 4, 8, 12, 16, 20, 24, 28, 32, 36\}$ of the 36-layer Qwen3-4B with contrastive analysis per Rinsky et al. [2024]; the three directions (commitment, reflection, peer-framing) are introduced in §5.

3 Behavioral comparison at matched compute

Across the harder-than-L3 evaluations, no condition exceeds both controls on more than one benchmark, and D’s adversarial robustness fingerprint is the single behavior no other condition reproduces in full.

Table 2: Pass@1 on the headline benchmarks. L4 and GPQA at $n=8$, $T=0.6$; L5 at $n=256$, $T=0.6$. Bold marks the best non-base condition per column.

	L4	L5	GPQA Diamond
base	0.672	0.552	0.444
S	0.646	0.527	0.480
D	0.742	0.605	0.549
C	0.758	0.620	0.573
DPP	0.750	0.651	0.420
CAP v2	0.662	0.597	0.364

3.1 Capability across benchmarks

C is the strongest condition on the two largest harder-than-L3 evaluations, beating every multi-agent variant and both single-agent conditions. D does not exceed its compute-matched control on accuracy. DPP overtakes C only on L5 (0.651 vs 0.620); the result is suggestive but limited to one benchmark. Pass@ k at $n=256$ on L5 shows D, C, DPP, and CAP v2 all exceed base at every k while S sits at or below base.

Two conditions underperform base on GPQA Diamond: DPP (-2.4pp) and CAP v2 (-8pp). Both concentrate gradient pressure on confident-correct math rollouts, specializing the policy at the cost of out-of-distribution transfer. The GPQA drop limits the claim that CAP v2 fully reproduces multi-agent training’s benefits but does not undermine its adversarial-robustness finding (§3.2).

3.2 Adversarial robustness

Table 3: Regressive flip rate (M12) on the include-yourself intersection and progressive flip rate (M13) on a 50-problem MATH-500 L5 subset. M12 denominators differ across conditions (§2.2); the construction is identical. Lower M12 means better adversarial robustness; higher M13 means the model integrates correct counter-arguments more often.

	M12 regressive flip	M13 progressive flip
S	0.070 (6/86)	0.391 (9/23)
D	0.012 (1/86)	0.250 (5/20)
C	0.047 (4/86)	0.364 (8/22)
D-prime	—	0.333 (6/18)
DPP	0.040 (4/99)	0.350 (7/20)
CAP v2	0.024 (2/85)	0.400 (8/20)

D’s adversarial flip rate is roughly $6\times$ lower than S’s and $4\times$ lower than C’s. This is not an accuracy gap; C and D are close on capability but differ by a factor of four on this measure. CAP v2 reaches 0.024, placing it between D and C and much closer to D: a single-agent training regime with no peer reproduces most of D’s adversarial-robustness fingerprint. The remaining factor-of-two gap between CAP v2 and D is the open question for §6.

Reflection markers. D produces meaningfully fewer surface reflection markers per rollout (“wait,” “actually,” “let me check,” “hmm,” “hold on”) than S or C at similar pass@1. D commits to its answers more firmly under counter-argument but verbalizes less of the reasoning that would explain that commitment. The mechanism analysis in §5 answers whether D reflects more internally without surfacing the markers, or reflects less, by extracting and steering on a reflection direction.

DPP M12 recovery and M13 worsening. DPP’s iter 75 $\rightarrow$ iter 100 trajectory shows opposite-direction movement on the two measures. M12 improves $0.062 \rightarrow 0.040$, partly retracting the earlier reading that the strict gate had eliminated D’s robustness. M13 simultaneously worsens $0.222 \rightarrow 0.350$: DPP iter 100 is more flippable to generic counter-arguments than D ever was. The disambiguating steering experiments are pending (§7); the defensible observation is that stubbornness is not a single thing.

3.3 Mutual-swap progression

The three multi-agent variants span a progression in persuader-gate strictness: D uses CRISP’s original loose gate, D-prime keeps it but adds heterogeneous Bob, KL anchors, a stability bonus, and an XOR-curriculum filter, and DPP additionally tightens the gate to require the persuader to hold position post-discussion.

Table 4: Mutual-swap progression. Mutual swap = both players flip their answers during discussion. Productive persuasion = one player flips toward the other’s correct answer while the persuader holds position. Rates over cumulative training disagreement events.

	Mutual swap	Productive persuasion	n disagreement events
D	47%	15%	111
D-prime	33.9%	~15%	121
DPP	30.6%	15.3%	180

Mutual swap tightens 47% $\rightarrow$ 30.6% while productive persuasion holds at 15%. DPP’s strict gate cuts overall bonus volume by roughly 5 $\times$; the volume-versus-purity tension this creates is the subject of §6.

4 Analytical controls

A solo-RLVR baseline trained on the same problems under the same algorithm differs from a multi-agent variant in two ways at once: it consumes less compute, and it never receives the gradient-shaping signal the multi-agent reward formula delivers. Both confounds need to be ruled out before the peer mechanism can be credited with the gain. The two controls in this section each isolate one of them. C holds compute equal while removing peer interaction. CAP v2 delivers the persuader bonus’s gradient signal in a single-agent setting. Each is the training-time analogue of the matched-compute argument worked through at inference time by Wang et al. [2024] and Becker et al. [2026].

4.1 Compute-matched control (C)

C keeps D’s full two-player infrastructure and disagreement detection (§2.1). On disagreement, instead of exchanging drafts, each player generates one additional rollout from a self-draft prompt structurally identical to D’s, with two solutions from the same player rather than one from each. The extra rollout joins the player’s training pool under standard GRPO. Per disagreement event the two conditions spend identical compute under the same prompt template; the only thing that differs is whether the extra compute is spent on peer interaction or on additional samples from the player’s own distribution.

The behavioral comparison in §3 delivers a sharper result than the methodology was designed to detect. C beats D on accuracy on the two largest harder-than-L3 evaluations (L4 0.758 vs 0.742; GPQA Diamond 0.573 vs 0.549). C does not reproduce D’s adversarial robustness (M12 0.047 vs D’s 0.012). Read together: the extra compute spent on hard problems accounts for the accuracy gain of multi-agent training over solo, and the discussion protocol’s distinctive contribution is the adversarial-robustness fingerprint, not the accuracy gain. C narrows the mechanism question to what the discussion protocol does that more sampling on the same problems does not.

4.2 Confidence-amplified control (CAP v2)

Motivation. The persuader bonus in D delivers a specific gradient signal: extra weight on the model’s pre-discussion correct rollouts when its peer adopts the answer. If D’s adversarial robustness comes from gradient pressure of that shape, peer interaction is not strictly needed to deliver it. CAP v2 tests this directly. It is solo training plus a per-rollout amplification of the policy-gradient contribution from correct rollouts on problems the model already solves confidently. No peer, no discussion stage, no two-player infrastructure.

CAP v1 and its failure. The natural first design (CAP v1) added the amplification at the reward layer: for a problem with $\geq 5/8$ correct rollouts, the reward of each correct rollout was shifted by $+\gamma$

with $\gamma = 0.3$. This is the most literal translation of "amplify the gradient signal on these rollouts." Training collapsed by iter 80. The no-box rate climbed from roughly 0.35 to 1.0, and pass@1 dropped from 0.74 at iter 25 to near zero at iter 100.

Diagnosis: GRPO normalizes within a group by per-group z-score, but the running baseline of expected reward magnitude across problems is computed on raw rewards. The bonus inflated this baseline, so correct rollouts on harder problems (where the bonus didn't fire) received negative advantages relative to the inflated baseline. The cheapest behavioral response was to abstain: the no-box penalty of -0.5 was less costly than wrong-with-box against the inflated baseline. The model learned to stop emitting boxed answers on hard problems, the behavior generalized across difficulty, and training collapsed. This is a structural property of adding reward to a subset of events without preserving baseline magnitude, not an implementation bug.

Redesign. CAP v2 moves the amplification from the reward layer to the loss layer. For a correct rollout i on a confident-correct problem, the two designs differ as follows: CAP v1: $r_i \leftarrow r_i + \gamma$, $A_i = (r_i - EMA)/\sigma_g$

CAPv2: r_i unchanged,

$A_i = (r_i - EMA)/\sigma_g$,

$A_i \leftarrow (1+\gamma) \cdot A_i$ The v1 update inflates the EMA baseline through the shifted r_i ; the v2 update leaves r_i and the baseline alone, amplifying only the gradient step on the targeted rollouts. Mathematically equivalent to weighting the policy-gradient contribution of those rollouts by 1.3, with no perturbation to the reward distribution.

CAP v2 trains stably through iter 100. It reaches $M12 = 0.024$ (compared to D's 0.012, C's 0.047, and S's 0.070) on 2/85 on the four-condition include-yourself intersection, reproducing most of D's adversarial-robustness fingerprint with no peer mechanism. The remaining factor-of-two gap between CAP v2 and D, alongside the three-way axis dissociation across D, DPP, and CAP v2 documented in §5, is what makes the volume hypothesis directly testable in §6.

5 Mechanism analysis

The behavioral fingerprint in §3 is underdetermined as a mechanism story; this section asks whether D's training reshapes the model's residual-stream representations in ways that explain D and CAP v2's adversarial robustness relative to C.

5.1 Axis extraction

We extract three contrastive directions per condition following Rimsky et al. [2024].

Commitment axis. The mean of residual-stream activations on correct rollouts minus the mean on wrong rollouts, taken at the last reasoning token before `\boxed{` opens. We follow Zhang et al. [2025] and initially label this the "correctness direction." The steering experiment in §5.2 shows it actually controls format compliance and revision rate, not correctness, so we relabel it the commitment axis. Peak layer L32.

Reflection axis. The mean of residual-stream activations at the predecessor of a reflection-trigger token ("wait," "actually," "let me check/reconsider/verify," "hmm," "hold on"), minus the mean at a matched non-trigger token from the same rollout. Probing the predecessor token rather than the trigger eliminates surface-form leakage. Follows Zhu et al. [2025]. Peak layer L36.

Peer-framing axis. The mean over a peer-framed counter-argument prompt ("Your peer Bob says...") minus the mean over a source-unspecified prompt ("Consider the following..."), at shared-suffix positions. The axis tests whether D's adversarial robustness comes from peer-sycophancy: if so, D should engage this axis more than S or C. D's L32 norm is 3.25, the lowest of any condition (vs base 3.47, S 3.57, C 4.45). The hypothesis is falsified by direct measurement. Peak layer L32.

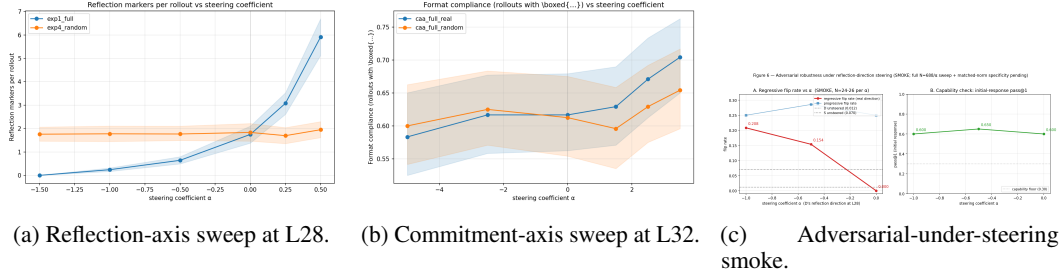


Figure 2: Steering on D. **(a):** Marker rate moves monotonically with α under the real direction (blue), 0.24 at $\alpha = -1$ to 5.91 at $\alpha = +0.5$; matched-norm random direction (orange) is flat at ≈ 1.8 . **(b):** Format compliance climbs with α under the real direction. The companion pass@1 plot (Appendix B) stays flat across the working range, with no advantage of the real direction over random — the basis for the commitment-versus-correctness reframe. **(c):** Regressive flip rate (red) moves $0.000 \rightarrow 0.154 \rightarrow 0.208$ across $\alpha \in \{0, -0.5, -1.0\}$; progressive flip rate (blue) flat across the same range. Smoke result; 5 problems $\times$ 8 rollouts per α .

5.2 Steering on D

Cross-condition geometry is correlational. Three activation-steering experiments convert the correlations into causal claims by adding $\alpha \cdot \text{direction}$ to the residual stream at a chosen layer during generation. Each includes a matched-norm random-direction control where available, and an $\alpha = 0$ hook sanity check.

Reflection-axis sweep at L28. We steer D using D’s reflection direction at L28, the layer where the direction is cleanest of unembedding-adjacency artifacts. Coefficients $\alpha \in \{-1.5, -1.0, -0.5, 0, +0.25, +0.5\}$; 30 MATH-500 L5 problems $\times$ 8 rollouts per (problem, α). Under the real direction, the rate of surface reflection markers per rollout moves from 0.24 at $\alpha = -1.0$ to 1.75 at $\alpha = 0$ to 5.91 at $\alpha = +0.5$, a roughly $24\times$ swing (Figure 2a). The matched-norm random direction stays flat at ≈ 1.8 markers per rollout across the full α range. The marker rate change is specific to this direction as random L28 perturbations of the same magnitude don’t move it. Capability is preserved across the working range: pass@1 stays in $[0.454, 0.558]$ across $\alpha \in [-1, +0.5]$. Only $\alpha = -1.5$ breaks generation (format compliance drops to 0.004).

Commitment-axis sweep at L32. We steer D using D’s commitment direction at L32, $\alpha \in \{-5, -2.5, 0, +1.5, +2.5, +3.5\}$. The wider range compensates for the commitment direction’s smaller norm. Format compliance climbs monotonically with α under the real direction: 0.583 at $\alpha = -5$, 0.617 at $\alpha = 0$, 0.704 at $\alpha = +3.5$. Revision rate spikes at $\alpha = +3.5$ to 0.113 under the real direction vs 0.021 under random. But pass@1 is flat-to-down across the entire sweep, with the random direction actually higher at $\alpha = +3.5$ (0.550 vs 0.483). The contrastive direction reads commitment cleanliness, not correctness, and the pass@1 correlation in the original contrastive labels comes from correct rollouts committing more cleanly to one answer than wrong rollouts do. Methodologically, CAA on rollout-level outcome labels can pick up something other than the labeled concept when within-problem label variance is missing; a behavioral steering check is a cheap way to catch this.

Adversarial-under-steering smoke. The headline causal question: does the reflection axis drive D’s adversarial robustness? We generate D’s initial answer under steering, append a wrong counter-argument, then generate the reconsidered answer with the hook still active. Five problems, 8 rollouts per $\alpha \in \{0, -0.5, -1.0\}$. Regressive flip rate moves $0 \rightarrow 0.154 \rightarrow 0.208$ (Figure 2c); at $\alpha = -1.0$, D is roughly $3\times$ more flip-prone than unsteered S. Progressive flip rate (accepting a correct counter-argument when initially wrong) stays flat near 0.25. Suppressing the reflection axis breaks resistance to wrong claims without changing general persuadability. This is a 5-problem smoke; the matched-norm specificity control on the adversarial sweep is pending (§7).

5.3 Three-condition dissociation

D, DPP, and CAP v2 all receive some form of confident-correct gradient signal: D via the loose persuader bonus, DPP via the strict persuader bonus, CAP v2 via the advantage multiplier. They diverge sharply on adversarial robustness: D at 0.012, CAP v2 at 0.024, DPP at 0.040. The mech-axis numbers explain why.

Table 5: Cross-condition mech-axis matrix at iter 100 (iter 75 for D-prime, not extracted). V2 norm and sanity at L32; V3 norm and cos-to-base at L36; V5 norm at L32. V2 sanity is the random-label ratio: ratios 0.30 pass, larger ratios are FLAGGED. Three of the six conditions are FLAGGED on V2 (base, D, CAP v2), three pass (S, C, DPP).

	V2 norm	V2 sanity	V3 norm	V3 cos to base	V5 norm
base	30.99	FLAG (0.42)	110.04	—	3.47
S	67.52	PASS (0.24)	115.49	+0.84	3.57
D	35.21	FLAG (0.39)	116.78	+0.98	3.25
C	55.10	PASS (0.28)	105.89	+0.91	4.45
DPP	64.08	PASS (0.19)	109.52	+0.96	3.40
CAP v2	29.53	FLAG (0.45)	114.52	+0.89	3.48

D has the largest V3 norm of any condition (116.78, +6.7 above base). Its V3 direction is strongly aligned with base (cos +0.978). Its V2 norm is base-like (35.21 vs base 30.99) and FLAGGED on the random-label sanity check (ratio 0.39). D’s commitment direction is statistically weak. This is consistent with the steering result in §5.2: the direction controls commitment cleanliness rather than correctness. D trains the reflection axis substantially. It does not cleanly train the commitment axis.

DPP (iter 100) inverts this pattern. Its V2 norm is the largest of any condition (64.08). Its V2 sanity ratio is 0.187, the cleanest of any condition. Its V3 norm sits at 109.52, essentially at base. DPP cleanly trains the commitment axis but barely moves the reflection axis. The strict persuader gate produces clean per-firing gradient signal. It also fires roughly 5× less often than D’s loose gate (§3.3). The cleanliness shows up on V2. The reduced volume shows up on V3.

CAP v2 trains the reflection axis without training the commitment axis. V3 norm reaches 114.52 with cos +0.888 to D’s reflection direction. That’s about 70% of D’s gain over base, in the same direction. V2 norm sits at 29.53, at base, FLAGGED on sanity (ratio 0.45). A single-agent training regime with no peer reproduces most of D’s reflection-axis training and none of the commitment-axis movement DPP achieves. Combined with $M12 = 0.024$, this confirms the load-bearing part of the volume hypothesis: peer interaction is not necessary to train the reflection axis to D-like magnitudes.

The three-way dissociation is the data on which §6 reads a volume-versus-purity tradeoff in reward design.

6 Discussion

What the gate strictness controls. The persuader bonus fires on the persuader’s pre-discussion correct rollout, pushing toward “commit confidently to your initial correct answer.” Gate strictness controls *when* this signal fires: D’s loose gate fires often (47% mutual-swap, 15% productive — §3.3), with most firings structurally pathological but the volume present. DPP’s strict gate fires roughly 5× less often, on clean events only. Same gradient direction, different volume and per-firing purity.

Why the axes train differently. The reflection axis needs gradient volume. D’s V3 norm reaches 116.78 (§5.3), the largest in the comparison set. DPP’s V3 sits at 109.52, essentially at base. The commitment axis benefits from per-firing purity. DPP’s clean firings trained V2 to 64.08 with the cleanest sanity ratio of any condition (0.187). D’s noisy gate left V2 FLAGGED on sanity (0.39) at norm 35.21. The two axes have separable training requirements, and the standard persuader-bonus formulation gates both with the same trigger. We call this the **volume-versus-purity tradeoff**: improving signal purity reduces firing volume, and vice versa. CAP v2 partially decouples them by moving the gradient amplification out of the persuader-bonus channel entirely.

CAP v2 substantially confirms the volume hypothesis. CAP v2 trains the reflection axis (V3 norm 114.52, $\cos +0.888$ with D’s direction, about 70% of D’s gain over base) without any peer mechanism. The caveat: CAP v2 underperforms base on GPQA Diamond (0.364 vs 0.444), so confidence amplification on math rollouts specializes the policy at the cost of out-of-distribution transfer.

Implications for evaluating multi-agent RLVR. Future work on multi-agent RLVR would benefit from reporting against the controls developed here: C isolates the protocol from the compute it consumes, CAP v2 isolates it from the gradient-shaping signal, and a protocol that beats both is doing work neither control captures. Each is a one-flag modification to an existing training pipeline. We claim only that these are stronger than the standard solo baseline for isolating specific mechanism candidates, not that they exhaust the space of useful comparisons.

7 Limitations

Single base model, single seed. The entire investigation runs on Qwen3-4B-Instruct-2507, and each condition was trained once. We do not characterize across-seed variance or replicate on a second base model, so the volume-versus-purity tradeoff and CAP v2’s partial confirmation are single-observation findings.

Design-family scope. Three multi-agent variants are tested, all CRISP-family protocols. External multi-agent RLVR systems (MAPoRL, Dr. MAS, Stronger-MAS) were not benchmarked, and a protocol with a structurally different reward shape could exceed both controls. Benchmarking external protocols against C and CAP v2 is a natural next step.

Steering on D only. The activation-steering experiments in §5.2 run on D; the DPP and CAP v2 norms reported in §5.3 come from contrastive direction extraction without accompanying steering. The three-condition dissociation is partly correlational across DPP and CAP v2: the directions exist, but whether perturbing them produces the predicted behavioral changes is untested for those conditions. The adversarial-under-steering smoke also runs on only 5 problems; the matched-norm specificity control on the full 86-problem sweep is queued.

Out-of-distribution generalization. CAP v2’s GPQA Diamond underperformance against base (§6) limits the volume hypothesis to the adversarial-robustness metric that motivated CAP v2, not to capability transfer broadly.

8 Related work

Multi-agent RLVR training. MAPoRL [Park et al., 2025], CRISP [Baumgartner and Agrawal, 2026], Dr. MAS [Feng et al., 2026], and Stronger-MAS [Zhao et al., 2026] train models with multi-agent discussion via RL, each evaluated against a solo-RLVR baseline that conflates peer interaction with the protocol’s extra compute and gradient-shaping. The CRISP family is the design space this paper investigates.

Matched-compute comparisons and reward shaping. Wang et al. [2024], Becker et al. [2026], and Wynn et al. [2025] find single-agent controls match or beat multi-agent protocols at matched compute at inference time; we ask the analogous question during training. The CAP v1 collapse and the volume-versus-purity tradeoff are two instances of reward-shaping interactions with GRPO/DAPO’s per-group advantage normalization [Guo et al., 2025, Yu et al., 2025] that are not extensively studied.

Linear representations and CAA. Rimsky et al. [2024] introduces the contrastive activation addition methodology used throughout, building on Marks and Tegmark [2024]’s observation that concepts encode as linear directions in residual streams. Arditi et al. [2024] is the closest precedent for our single-axis causal claim; Zhu et al. [2025] and Zhang et al. [2025] provide the reflection and commitment axis extraction templates. Betley et al. [2026] and Chen et al. [2025] apply a similar matched-conditions probing template to narrow-fine-tuning misalignment.

9 Conclusion

At matched compute, the discussion protocol’s distinctive contribution is adversarial robustness, traced to a volume-versus-purity tradeoff in reward design that CAP v2 partially circumvents through single-agent gradient amplification.

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A Hyperparameters and training details

Table 6 lists training hyperparameters shared across all trained conditions, with footnotes for condition-specific differences.

Table 6: Training hyperparameters. Matched across all trained conditions (S, D, C, DPP, CAP v2) except where noted; condition-specific deviations are flagged inline.

Parameter	Value
Base model	Qwen3-4B-Instruct-2507 (36 layers, hidden 2560, bf16)
GRPO algorithm	DCPO-style asymmetric-clip GRPO
Rollouts per problem per player	8
Problems per batch	8
Optimizer	DeepSpeed FusedAdam ($\beta_1=0.9$, $\beta_2=0.999$), wd 0.01
Peak learning rate	1×10^{-6}
LR warmup	Linear, 20 iters
LR decay (post-warmup)	Cosine to $0.1 \times$ peak by iter 200
Clip ratio ϵ (low / high)	0.2 / 0.28
DCPO α (importance-ratio scaling)	3.0
KL coefficient (forward JS vs frozen base)	0.005
Sampling temperature, Alice	0.6
Sampling temperature, Bob (D, C only)	1.0
Max sequence length	12,288 tokens
Training data	DAP0-Math-17K (HuggingFace <code>open-r1/DAP0-Math-17k-Processed</code>)
Curriculum filter	2,023 problems, solve-rate band $[0.25, 0.75]$ at $T=0.6$
Total iterations	200
Hardware	$2 \times$ H200, DeepSpeed ZeRO-2, AdamW offload
Task reward	SymPy-verified binary correctness on final <code>\boxed{\}</code>
No-box penalty	-0.5
Persuader bonus γ_1 (D only)	$+0.3$ additive
Compute-control mode (C only)	<code>self_draft</code> , 1 extra rollout per player per disagreed problem

Persuader bonus formula (D only). A player p is the *persuader* for a given problem if: (i) the majority pre-discussion answer of p is correct, (ii) the majority pre-discussion answer of the peer $\neg p$ is incorrect, and (iii) the majority post-discussion answer of $\neg p$ is correct. When these three conditions all fire, every correct pre-discussion rollout of p receives an additive bonus $\gamma_1 = 0.3$ on top of its base task reward. No bonus is applied in any other case (mutual swap, both correct, both wrong, or peer didn’t flip). The bonus is additive, not multiplicative. DPP uses the same gate plus a fourth condition: the persuader’s own majority post-discussion answer must remain correct (§2.1).

Compute-matching protocol for C (self-draft). On each disagreed problem (at least 2 distinct boxed answers among parseable rollouts), each player generates one additional rollout from a self-draft prompt structured identically to D’s discussion prompt, except the two solutions shown are both from the same player (selected via preference for both-correct or both-wrong pairs). The extra rollout joins the player’s training pool for that problem and receives standard GRPO advantage and gradient treatment. The compute spent matches D’s discussion-stage compute exactly: 1 extra rollout per player per disagreed problem, same prompt structure, no peer signal.

B Supplementary figures

B.1 Cross-condition geometry across layers

B.2 Reflection axis

B.3 Peer-framing axis

B.4 Commitment axis

Figure 2 — Cross-condition cosine to base across layers, three axes

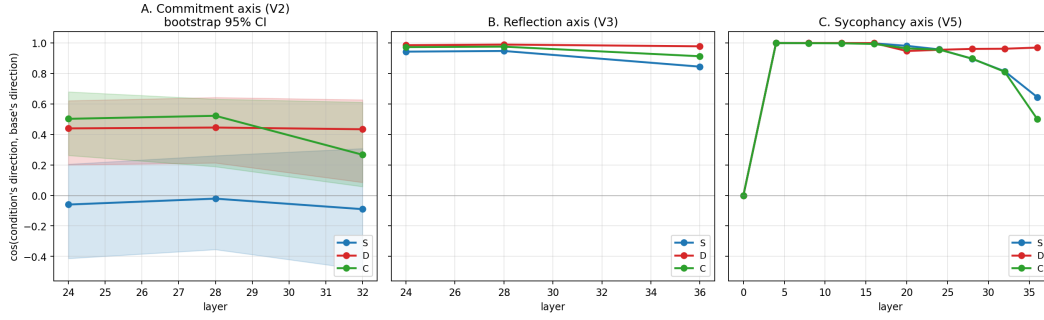


Figure 3: Cross-condition cosine to base across layers for the three contrastive axes. **A:** Commitment axis (V2) at L24/28/32 with bootstrap 95% CI ribbons. D and C are positive throughout; S sits near zero. **B:** Reflection axis (V3) at L24/28/36. All three trained conditions strongly aligned with base (0.84–0.99). **C:** Peer-framing axis (V5) across all 10 probed layers. Conditions cluster tightly through mid layers; S diverges at L36. Cross-condition pattern is layer-stable, not an artifact of the peak-layer choice in §5.1.

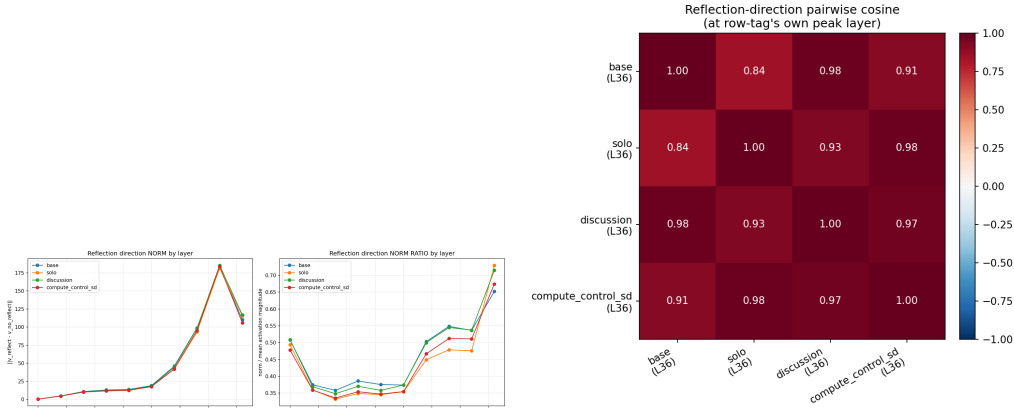


Figure 4: Reflection direction. **Left (S1):** L2 norm by layer for all four conditions. Norm grows monotonically and peaks at L36 with values within 10% of each other (base 110.0, S 115.5, D 116.8, C 105.9). **Right (S2):** pairwise cosine of the reflection direction at each model's L36 peak. All trained conditions remain highly aligned with base (S 0.844, D 0.978, C 0.913) and with each other (0.91–0.98). Reflection direction preserves orientation across training; the cross-condition difference lives in typical engagement (Figure 3), not in direction geometry.

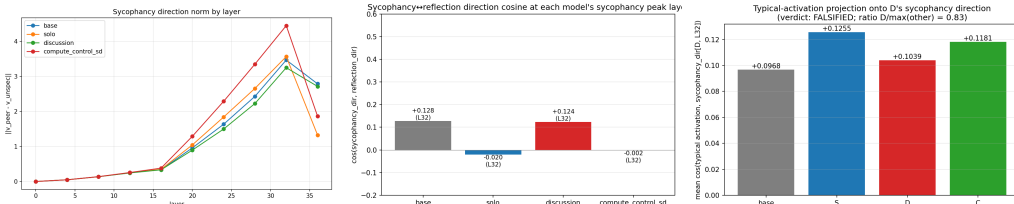


Figure 5: Peer-framing axis. **Left (S3):** direction norm by layer. L0 norm is exactly zero by construction (matched-token boundaries), a hard sanity check. Peak at L32: base 3.47, S 3.57, D 3.25, C 4.45. D has the *smallest* norm of the four conditions, opposite the prediction of the peer-sycophancy hypothesis. **Center (S4):** cosine between each condition's peer-framing and reflection directions at L32 (base +0.128, S -0.020, D +0.124, C -0.002). All near zero; the two axes are essentially independent. **Right (S5):** typical-activation projection onto D's L32 peer-framing direction. Base +0.097, D +0.104, C +0.118, S +0.126. D engages this axis less than C or S, falsifying the peer-sycophancy hypothesis ($D/\max = 0.83$, below the supported threshold).

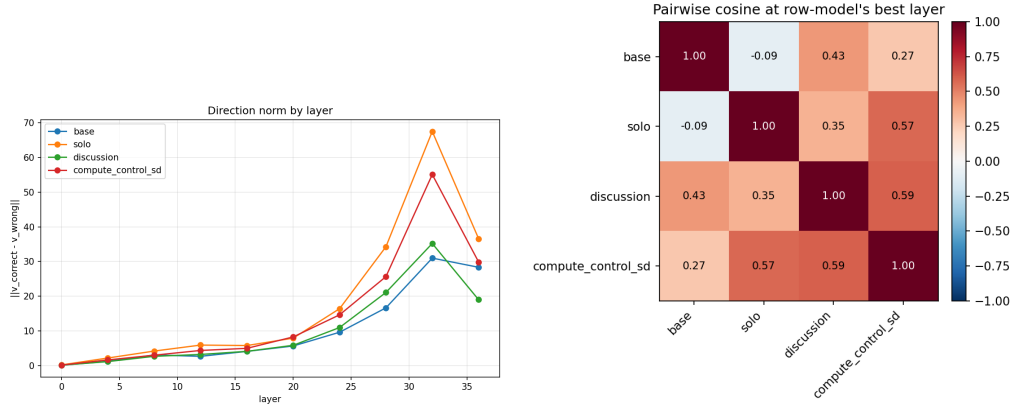


Figure 6: Commitment direction. **Left (S6):** direction norm by layer. Peak at L32: base 30.99, S 67.52, D 35.21, C 55.10. S has the largest commitment direction (norm-ratio 0.207 vs base 0.096, D 0.111, C 0.171), consistent with strong gradient pressure to distinguish correct from wrong commits during solo RL. D’s norm is base-like. **Right (S7):** pairwise cosine at L32. D-to-base +0.434 and C-to-base +0.267 are positive; S-to-base is -0.090 . S sits in a different commitment-axis orientation than base while D and C preserve alignment. This is the geometric signature behind the cross-condition pattern in Figure 2A.